

Assignment B:

Motion assessment of a semisubmersible in waves



Introduction

A geological survey of the sea floor in a remote deep-water is planned. It is not possible to set up a fixed installation. Instead, the base installation is a semisubmersible (“semisub”) from where the survey is controlled and practically conducted.

In this assignment, a semisub must be assessed for its ability to be used for the geological survey.¹ The main particulars and geometrical details of the semisub are given in Appendix A. The assessment must be based on the hydrodynamic synthesis by Paulling (1985), and the (undamped) heave motion is studied exclusively, neglecting coupling to other motion components.

Deadline for the report: 13th May 2026 at 23:59. Be sure to read the ‘Practical remarks...’ on page 3.

Information about the sea state

The ability of the semisub to operate is assumed to depend solely on the sea state, noticing that operational activities must be stopped if the platform’s heave motion is larger than a certain threshold level. The semisub must be assessed for a sea state described by a Bretschneider spectrum. Long-crested waves are assumed throughout, and a dynamic positioning system ensures that the semisub is always aligned with the x-axis parallel to the wave crests, see Figures 1 and 2.

Questions

In addition to the information given in the preceding and in the questions themselves, any other necessary information and data must be consulted from the literature or by reasonable assumptions. Whenever relevant, the discrete set of wave frequencies must be based on a lower cut-off frequency corresponding

¹ The assessment does not concern the safety of the actual design and its structural integrity; we are just assessing if operational activities can be conducted.

to $\omega_{\text{low}} = 0.01$ rad/s and an upper cut-off frequency corresponding to $\omega_{\text{high}} = 2\pi$ rad/s, and, in between, the frequencies are spaced by 0.001 rad/s. That is, the wave frequency vector is formed by (0.01:0.001:2 π) rad/s.

Part A: Sea state

1. We consider conditions during a storm, lasting three hours, where the sea state is characterized by significant wave height $H_s = 5.5$ m and zero-upcrossing period $T_z = 10$ s. Simulate one likely realization (a *sample*) of observed wave heights during the storm; the individual wave heights follow the Rayleigh distribution, i.e., $P(H \leq h) \equiv F(h) = 1 - \exp\left(-2\left(\frac{h}{H_s}\right)^2\right)$. Show a histogram of the realized wave heights, i.e., show the number of occurrences at given heights. What is the maximum wave height H_{max} in the sample?
2. Repeat 1 and simulate the possible outcomes of H_{max} from 10,000 realizations of the storm and the resulting sea state (characterized by the same H_s and T_z). Plot the (“empirical”) probability density function of *all* observed H_{max} . Make a fit to the data with an *extreme value distribution*. (Hint: In MATLAB, consult the function `fitdist`)
3. Based on the fit from Q2, what is the most probable largest (MPL) $H_{\text{max,MPL}}$ that would occur if 10,000 of such storms had occurred? Compare this result to the corresponding number derived by Ochi (1973).
4. From the plot in Q2, determine the chance that a *maximum* wave height larger than $H_{\text{max,MPL}}$ occurs. Is it consistent with theory (Ochi, 1973)?
5. Plot the wave spectrum corresponding to the parameters in Q1. NB: The horizontal axis must show the frequency in Hz; with the y-axis scaled accordingly.
6. Compute (by numerical integration) the spectral moments m_0 , m_1 , m_2 , and determine from these the significant wave height and, respectively, the peak period, the mean period, and the zero-upcrossing period of the wave system. What are your comments?

Part B: Assessment of the motion response

Hint: *Before* starting to answer the questions, read Chapter 15 in Nielsen (2024).

7. What is the platform’s mass [tons]? Compute also the *effective* (“total”) mass of the semisub, including the hydrodynamic, *added* mass? Note, in computing the added mass you must consult Appendix B (which is taken from DS449, 1983).
8. Write down the equation of uncoupled and undamped heave motion.
9. Determine the natural (undamped) period of the heave motion of the semisub.
10. Following the synthesis outlined by Paulling (1985), solve the equation of heave motion in terms of the amplitude response (RAO) of the transfer function. In doing this, you must present the expressions forming the solution², including the expressions for the wave excitation forces. Plot the RAO.
11. Plot the resulting response spectrum when the semisub works in the sea state described by the given wave spectrum, as determined in Questions 1 and 5.

² Note that the platform studied herein is different from the one studied by Paulling (1985) and the one studied in the lecture notes; see also the middle paragraph on page 208 in the lecture notes (Nielsen, 2024).

12. Compute statistics that can be used to quantify the operability of the semisub; this should include the calculation of the most probable largest (MPL) heave amplitude h_{MPL} in a 2-hours period, cf. Ochi (1973). How many exceedances of the level h_{MPL} are expected during two hours?

References

DS449 (1983). *Norm for pælefunderede offshore stålkonstruktioner*. Dansk Standard. [The standard has been replaced by DS/EN ISO 19902:2020, but this is not relevant herein.]

Nielsen, U.D. (2024). *Analysis of ship and other floating structures*, lectures notes (DTU Construct).

Ochi, M. (1973). *On prediction of extreme values*, Journal of Ship Research, Vol. 17, pp. 29-37. [The paper is available from DTU Learn. Note that Ochi uses the term most *probable extreme value* for MPL.]

Paulling, J.R. (1985). *Hydrodynamic synthesis of marine structures*, IUTAM, Proc. 16th Congress, Kgs. Lyngby, DK.

Practical remarks about the report

The front page of your report must inform names and study numbers of all group members. The list of content must indicate who is responsible for what.

Please also note that the report must be submitted in a PDF file format that allows text commenting (i.e. do NOT print your report with the pages as images). Include all necessary Matlab/Python scripts as prints in appendices. Finally, be sure to include your group number in the file name before submitting; e.g., Assign_B_Group_<insert number>.pdf.

Appendix A: Geometrical details and main particulars

As a general comment, it is noteworthy that the pontoons of a semisubmersible carry a substantial amount of ballast; this helps to ensure stability.

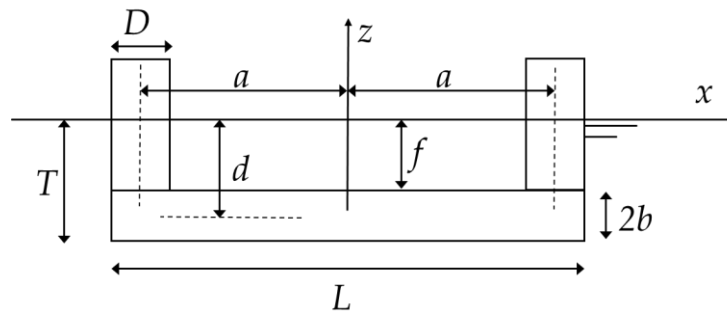


Figure 1: Sideview of the semisub; the y-z plane is identical. Note that the sketch is not in scale.

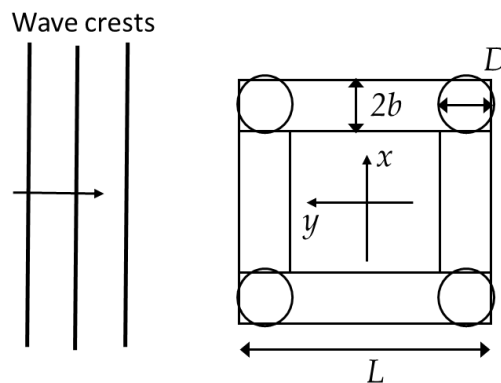


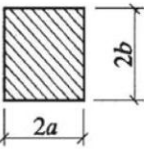
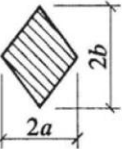
Figure 2: Topview of platform; the platform is double-symmetric.

Table 1: Main dimensions of the semisub.

$T = 18.5$ m (draught)
$L = 2a + D = 72.3$ m (length)
$D = 11.3$ m (diameter of columns)
$a = 30.5$ m (horizontal distance from platform centerlines to column centres)
$b = 5.40$ m (half pontoon breadth, half pontoon height)
$d = 13.1$ m = $T - b$
$f = 7.70$ m = $T - 2b$

Appendix B: Hydrodynamic mass coefficients for typical 2D geometries

Table 2: Relevant information for added mass calculations. From DS 449:1983.

Cross section	Direction of movement	$\frac{a}{b}$	C_m	A_R
	 Vertical		1,0	πa^2
	 Vertical		1,0	πa^2
	 Vertical		1,0	πa^2
	 Vertical		1,0	πa^2
	 Vertical	∞	1,00	πa^2
		10,0	1,14	
		5,0	1,21	
		2,0	1,36	
		1,0	1,51	
		0,5	1,70	
		0,2	1,98	
		0,1	2,23	
	 Vertical	2,0	0,85	πa^2
		1,0	0,76	
		0,5	0,67	
		0,2	0,61	